Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

Rate of adsorption of gas is greater when pressure increased

(A) For low range of pressure

(B) For high range of pressure

(C) For moderate range of pressure

(D) For every range of pressure

Q.2

Plot of log $\left(\frac{x}{m}\right)$ against. log P' is a straight. line inclined at an angle of 45°. When the pressure is 0.5 atm and Freundlich parameter, k is 10 . the amount of solute adsorbed per gram of adsorbent will be : $(\log 5 = 0.6990)$

(A) lg

(B) 2 g

(C) 3 g

(D) 5 g

Q.3

In an experiment, addition of 4.0ml of $0.005\text{M}\,BaCl_2$ to 16.0ml of arsenius sulphide sol. just causes the complete coagulation in 2 hrs. The flocculating value of the effective ion is -

(A) 0.75

(B) 2.0

(C) 1.0

(D) 0.5

Q.4

40ml of 0.8 N oxalic acid dihydrate is shaken with 0.75 g active charcoal. The final concentration of the solution after adsorption is 0.2 N. What is the amount of oxalic acid absorbed per gm of carbon?

(A) 3.15 g

(B) 3.45 g

(C) 6.30 g

(D) 2.02 g

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Q.5

One gram of activated charcoal has a surface area of $1000\ \mathrm{m}^2$. If complete coverage is assumed, how much ammonia (in cm^3 at NTP) could be adsorbed on the surface of $25\ \mathrm{g}$ of the charcoal ? Given : Diameter of $\mathrm{NH_3}$ molecule $= 0.3\ \mathrm{nm}$.

(A) $10148.8\ \mathrm{cm}^3$

(B) $11148.8\ \mathrm{cm}^3$

(C) $12148.8\ \mathrm{cm}^3$

(D) $13148.8\ \mathrm{cm}^3$

Q.6

Fixed parts of a colloidal solution of AgI are respectively $[\mathrm{AgI}]\mathrm{I}^-$ and $[\mathrm{AgI}]\mathrm{Ag}^+$ in presence of

(A) KI and $\mathrm{AgNO_3}$

(B) $\mathrm{AgNO_3}$ and KI

(C) KI and $\mathrm{KIO_3}$

(D) $\mathrm{AgNO_3}$ and $\mathrm{Ba(NO_3)_2}$

Q.7

In the titration of Cl^- ions in solution with $\mathrm{AgNO_3}$, sodium salt of fluorescein is used as an indicator. The equivalence point is marked with the appearance of reddish pink colour. This is due to -

(A) adsorption of Cl^- ions by AgCl precipitate particles

(B) adsorption of $\mathrm{NO_3}$ - ions by AgCl precipitate particles

(C) adsorption of Ag^+ ions by AgCl particle followed by adsorption of fluorescein anion

(D) reaction of fluorescein anion with Ag^+ ions in the solution

Q.8

Methylene blue, from its aqueous solution, is adsorbed on activated charcoal at $25^\circ\mathrm{C}$. For this process, the correct statement is

(A) The adsorption requires activation at $25^\circ\mathrm{C}$

(B) The adsorption is accompanied by a decrease in enthalpy

(C) The adsorption increases with increase of temperature

(D) The adsorption is irreversible

Q.9

During electrosmosis of $\mathrm{Fe(OH)_3}$ sol -

(A) sol particles move towards anode

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(B) sol particles move towards cathode

(C) the dispersion medium moves towards anode

(D) the sol particles do not move in either direction

Q.10

Which of the following statements are correct ?

(A) $\frac{k_P}{k_a} = e^{\Delta E/RT}$ where k_P = Rate constant for reaction with catalyst.
$\mathrm{K_a}$ = Rate constant for reaction without catalyst
$\Delta \mathrm{E}$ = Difference in activation energy for the two ways of reactions.

(B) Alcohol acts as a negative catalyst for the conversion of $\mathrm{Na_2SO_3} \longrightarrow \mathrm{Na_2SO_4}$.

(C) TEL (tetraethyl lead) acts as a negative catalyst in the knocking of petrol.

(D) Dilute $\mathrm{H_2SO_4}$ acts as a positive catalyst in the hydrolysis of esters to carboxylic acids and alcohols.

Q.11

Which of the following are correct statements

(A) Spontaneous adsorption of gases on solid surface is an exothermic process as entropy decreases during adsorption

(B) Formation of micelles takes place when temperature is below Kraft Temperature (T_k) and concentration is above critical micelle concentration (CMC)

(C) A colloid of $\mathrm{Fe(OH)_3}$ is prepared by adding a little excess (required to completely precipitate $\mathrm{Fe^{3+}}$ ions as $\mathrm{Fe(OH)_3}$) of NaOH in $\mathrm{FeCl_3}$ solution the particles of this sol will more towards cathode during electrophoresis.

(D) According to Hardy-Schulze rules the coagulation (flocculating) value of $\mathrm{Fe^{3+}}$ ion will be more than $\mathrm{Ba^{2+}}$ or $\mathrm{Na^{+}}$.

Q.12

Passage

For any analysis to achieve the desired level of accuracy, a calibration curve is usually prepared using concentrations of the standard which covers a reasonable range of analyte concentrations in a solution whose overall composition approximates that of the test solution. In real life, this is very near impossible and most analysts tend to rely on a procedure called "standard addition" in which a known quantity of the analyte is added to an aliquot of the sample. This procedure was applied to the determination of the concentration of phosphate in a patient's urine. A $5.0\ \mathrm{mL}$ urine sample was diluted to $100\ \mathrm{mL}$. The absorbance of a $25\ \mathrm{mL}$ aliquot of the solution was measured spectrophotometrically and found to be 0.428. A $1.0\ \mathrm{mL}$ sample of a solution containing $0.050\mathrm{mg}$ of phosphate was added to a second $25.0\ \mathrm{mL}$ aliquot and its absorbance determined to be 0.517.

Question

What is the absorbance due to the added phosphate ?

(A) 0.110

(B) 0.538

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(C) 0.310

(D) 0.210

Q.13

Passage

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Question

What is the concentration of phosphate in the aliquot of the specimen?

(A) $0.256\mathrm{mg}$

(B) $0.195\mathrm{mg}$

(C) $0.556\mathrm{mg}$

(D) $0.156\mathrm{mg}$

Q.14

Passage

For any analysis to achieve the desired level of accuracy, a calibration curve is usually prepared using concentrations of the standard which covers a reasonable range of analyte concentrations in a solution whose overall composition approximates that of the test solution. In real life, this is very near impossible and most analysts tend to rely on a procedure called "standard addition" in which a known quantity of the analyte is added to an aliquot of the sample. This procedure was applied to the determination of the concentration of phosphate in a patient's urine. A $5.0\ \mathrm{mL}$ urine sample was diluted to $100\ \mathrm{mL}$. The absorbance of a $25\ \mathrm{mL}$ aliquot of the solution was measured spectrophotometrically and found to be 0.428. A $1.0\ \mathrm{mL}$ sample of a solution containing $0.050\mathrm{mg}$ of phosphate was added to a second $25.0\ \mathrm{mL}$ aliquot and its absorbance determined to be 0.517.

Question

What is the concentration of phosphate in the patient's urine?

(A) $0.256\mathrm{mg/mL}$

(B) $0.356\mathrm{mg/mL}$

(C) $0.556\mathrm{mg/mL}$

(D) $0.156\mathrm{mg/mL}$

Q.15

Passage

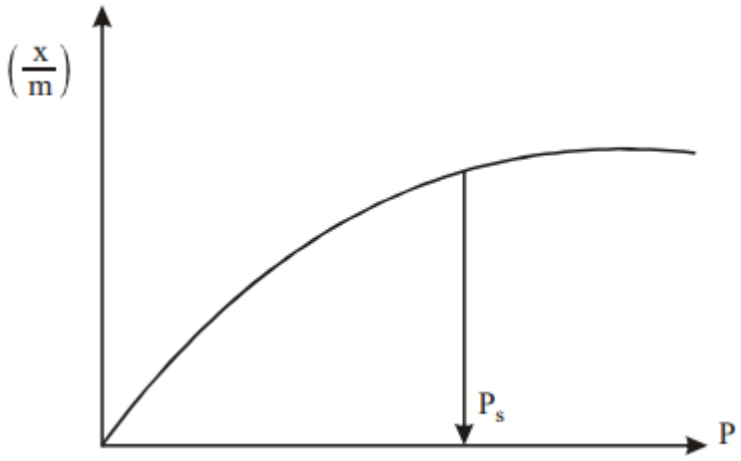
The physical adsorption is reversible. At a given temperature, the extent of adsorption will increase with the increase

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of pressure. It has been found that beyond certain pressure called saturation pressure, the extent of adsorption remains constant. At low pressures, the extent of adsorption $\left(\frac{x}{m}\right) \propto P$. At intermediate pressure $\left(\frac{x}{m}\right) \propto P^{1/n}$. At higher pressure (x/m) is independent of pressure. Freundlich adsorption isotherm was therefore written as $\left(\frac{x}{m}\right) = kP1/n$ where k is a constant. x is amount of adsorbent, and m is amount of adsorbate.

Question

The Freundlich isotherm is represented by the following curve.



Pick up the correct statement.

(A) $\left(\frac{x}{m}\right) = kP$ at high pressure

(B) $\left(\frac{x}{m}\right) = kP^{\circ}$ at high pressure

(C) $\left(\frac{x}{m}\right) = kP^{1/n}$ at very low pressure

(D) $\left(\frac{x}{m}\right) = k$ at moderate pressure

Q.16

Passage

The physical adsorption is reversible. At a given temperature, the extent of adsorption will increase with the increase of pressure. It has been found that beyond certain pressure called saturation pressure, the extent of adsorption remains constant. At low pressures, the extent of adsorption $\left(\frac{x}{m}\right) \propto P$. At intermediate pressure $\left(\frac{x}{m}\right) \propto P^{1/n}$. At higher pressure (x/m) is independent of pressure. Freundlich adsorption isotherm was therefore written as $\left(\frac{x}{m}\right) = kP1/n$ where k is a constant. x is amount of adsorbent, and m is amount of adsorbate.

Question

The following data were obtained for adsorption of carbon monoxide on 3 g of charcoal at $0^{\circ}C$ reduced to STP.

Pressure P mm	180	540
Volume of gas adsorbed (c.c)	16.5	38.1

Value of n is:

(A) 0.131

(B) 13.1

(C) 1.00

(D) 1.31

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Q.17

Passage

The physical adsorption is reversible. At a given temperature, the extent of adsorption will increase with the increase of pressure. It has been found that beyond certain pressure called saturation pressure, the extent of adsorption remains constant. At low pressures, the extent of adsorption $\left(\frac{x}{m}\right) \propto P$. At intermediate pressure $\left(\frac{x}{m}\right) \propto P^{1/n}$. At higher pressure (x/m) is independent of pressure. Freundlich adsorption isotherm was therefore written as $\left(\frac{x}{m}\right) = kP1/n$ where k is a constant. x is amount of adsorbent, and m is amount of adsorbate.

Question

Calculate the value of k in the Freundlich equation. Use data given in previous question

(A) $0.1044 ccg^{-1}$

(B) $1.044 ccg^{-1}$

(C) $10.44 ccg^{-1}$

(D) $1.044 \times 10^{-2} ccg^{-1}$

Q.18

Match the column

Column I	Column II
(A) Solution	(p) homogeneous
(B) Colloid	(q) heterogeneous
(C) Suspension	(r) Tyndall effect
(D) Optical property	(s) Gravitational force

(A) $(A) - p, (B) - q, r, (C) - q, s, (D) - r$

(B) $(A) - s, p, (B) - p, (C) - s, (D) - q$

(C) $(A) - q, r (B) - q, r (C) - p, q, r, s, (D) - p, s$

(D) $(A) - p, r, (B) - q, r, (C) - q, r, (D) - q, r$

Q.19

$100\ mL$ of $0.6M$ acetic acid is shaken with $2\ g$ activated carbon. The final conc. of the solution after adsorption is $0.5M$. The amount of acetic acid adsorbed per gram of carbon is $(x/10) g$. Find the value of x.

Q.20

10% sites of catalyst bed have absorbed by H_2. On heating H_2 gas is evolved from sites and collected at $0.03\ atm$ and $300\ K$ in a small vessel of $2.46\ cm^3$.
No. of sites available is 6.0×10^{15} per cm^2 and surface area is $1000\ cm^2$. Find out the no. of surface sites occupied per molecule of H_2. (Given $N_A = 6 \times 10^{23}$)

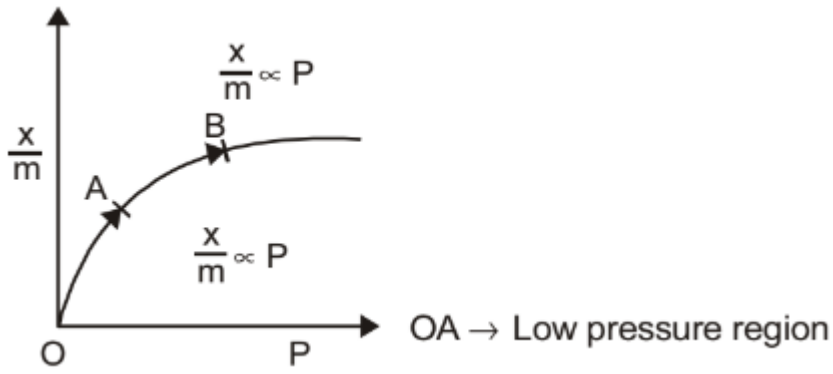
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Answers & Solutions

Q.1 Answer:

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Solution:



It is clear from the above graph that adsorption of gases is greater when pressure is increased in low range of pressure.

Q.2 Answer:

5 g

Solution:

According to Freundlich adsorption isotherm :

$\log\left(\frac{x}{m}\right) = \log k + \frac{1}{n}\log P$

When $\log\left(\frac{\mathrm{x}}{\mathrm{m}}\right)$ **is plotted against. log** P**. we get straight line of slope** $(1/\mathrm{n})$ **and intereopt** $(\log \mathrm{k})$**.**

$$\frac{1}{n} = \tan 45^\circ = 1$$

$$\log k = \log 10 = 1$$

$$\frac{x}{m} = k(P)^{1/n} = 10(0.5)' = 5$$

When $\mathrm{m} = 1\ \mathrm{g}, \mathrm{x} = 5\ \mathrm{g}$

Q.3 Answer:

1.0

Solution:

$\mathrm{As}_2\ \mathrm{S}_3$ **sol is negatively charged owing to preferential adsorption of** S^{2-} **ions. Cation would be the effective ion in coagulation.**

Flocculating value = minimum milli mol of the effective ion per litre of sol

$= \frac{4\times 0.005\times 10^3}{4+16} = 1.0$

Q.4 Answer:

2.02 g

Solution:

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Initial millimoles of acid $= 40 \times 0.4 = 16$ **millimole**
millimole left after adsorption $= 40 \times 0.1 = 4$ **millimole**
millimoles gets adsorbed $= 16 - 4 = 12$
So 0.75 **g charcoal adsorbs** $= 12 \times \frac{3}{4} = \frac{48}{3}$ **millimoles of acid**
$= \frac{48 \times 10^{-3} \times 126}{3}$ g $= 2.016$ g

Q.5 Answer:

13148.8 cm^3

Solution:

Total surface area available $= \left(25 \times 10^3\right) \text{m}^2 = 2.5 \times 10^8 \text{ cm}^2$
Surface area covered by one molecule of $\text{NH}_3 = \pi \text{r}^2$
$= \left[\frac{22}{7} \times \left(\frac{0.3}{2} \times 10^{-7}\right)^2\right] \text{cm}^2 = \left(\frac{22}{7} \times 0.15 \times 10^{-14} \times 0.15\right) \text{cm}^2$
$= 0.0707 \times 10^{-14} \text{ cm}^2 = 7.07 \times 10^{-16} \text{ cm}^2$
No. of ammonia molecules $= \frac{2.5 \times 10^8}{7.07 \times 10^{-16}} = 3.536 \times 10^{23}$
No. of moles of $\text{NH}_3 = \frac{3.536 \times 10^{23}}{6.023 \times 10^{23}} = 0.587$
Volume of NH_3 **at** $\text{NTP} = (0.587 \times 22400) = 13148.8 \text{ cm}^3$

Q.6 Answer:

KI and AgNO_3

Solution:

$$\text{AgI} + \text{KI} \rightarrow [\text{AgI}]\text{I}^- + \text{K}^+$$
$$\text{AgI} + \text{AgNO}_3 \rightarrow [\text{AgI}]\text{Ag}^+ + \text{NO}_3^-$$

Fixed part is $[\text{AgI}]\text{I}^-$ **and** $[\text{AgI}]\text{Ag}^+$

Q.7 Answer:

adsorption of Ag^+ ions by AgCl particle followed by adsorption of fluorescein anion

Solution:

After equivalence point, the AgCl **particles adsorb** Ag^+ **ions (present in slight excess) and then fluorescein anion. The precipitate particles turn reddish pink due to the complex formation between** Ag^+ **and fluorescein anion.**

Q.8 Answer:

The adsorption is accompanied by a decrease in enthalpy

Solution:

The adsorption of methylene blue on activated charcoal is physiosorption which is exothermic, multilayer and does not have energy barrier.

Q.9 Answer:

the dispersion medium moves towards anode

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Solution:

In electrosmosis, the sol particles are prevented from migration whereas the dispersion medium migrates in the direction opposite to those of particles. Here, the medium is negatively charged.

Q.10 Answer:

$\frac{k_P}{k_a} = e^{\Delta E/RT}$ where k_P = Rate constant for reaction with catalyst.

$\mathrm{K_a}$ = Rate constant for reaction without catalyst

$\Delta \mathrm{E}$ = Difference in activation energy for the two ways of reactions.

Solution:

1)TEL act as a negative catalyst. It is an anti-knocking substance is added to petrol to decrease the ignition of petrol vapours

2)Alcohol acts as a negative catalyst in the oxidation of Sodium sulfide

4)Esterification process is a slow process when it is performed at room temperature also the process does not complete at all in the room temperature. So to complete the process of esterification there should be something to complete to the process as well as to speed up the process. Thus concentrated sulphuric acid is added. It is a catalyst which will speed up the process and esterification will proceed speedily and complete on time.

Q.11 Answer:

Spontaneous adsorption of gases on solid surface is an exothermic process as entropy decreases during adsorption

Solution:

(A) $\Delta \mathrm{G} = \Delta \mathrm{H} - \mathrm{T}\Delta \mathrm{S} < \mathrm{O}$ **as** $\Delta \mathrm{S} < \mathrm{O}$ **so** $\Delta \mathrm{H}$ **has to be negative**

(B) Micelles formation will take place above $\mathrm{T_k}$ **and above** CMC

(C) This solution will be negatively charged.

(D) $\mathrm{Fe^{3+}}$ **ions will have greater flocculatibility power so smaller flocculating value.**

Q.12 Answer:

0.110

Solution:

First, correcting for the dilution of the aliquot during the second measurement.

Corrected absorbance $= \frac{0.571 \times 26.0}{25.0} = 0.538$

Absorbance due to 0.05 **mg of phosphate** $= 0.538 - 0.428 = 0.110$ mg

Q.13 Answer:

$0.195\mathrm{mg}$

Solution:

measurement.

Corrected absorbance $= \frac{0.571 \times 26.0}{25.0} = 0.538$

Absorbance due to 0.05 **mg of phosphate** $= 0.538 - 0.428 = 0.110$

mg **phosphate in the aliquot of specimen** $= \frac{0.428}{0.110} \times 0.05 = 0.195$ mg

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Q.14 Answer:

$0.156 \mathrm{mg/mL}$

Solution:

measurement.

Corrected absorbance $= \frac{0.571 \times 26.0}{25.0} = 0.538$

Absorbance due to 0.05 **mg of phosphate** $= 0.538 - 0.428 = 0.110$

mg **phosphate in the aliquot of specimen** $= \frac{0.428}{0.110} \times 0.05 = 0.195$ mg

Concentration of phosphate in the patient's urine $= \frac{100}{25.0} \times 0.195 \times \frac{1}{5.00} = 0.156 \mathrm{~mg/mL}$

Q.15 Answer:

$\left(\frac{\mathrm{x}}{\mathrm{m}}\right) = \mathrm{kP}^{\circ}$ at high pressure

Solution:

(i) The correct statement regarding Freundlich isotherm regarding physical adsorption is $\left(\frac{x}{m}\right) = kP^{\circ}$ **at high pressure.**

This is because at high pressure, the entire area of the adsorbate is completely covered by adsorbent.

Q.16 Answer:

1.31

Solution:

(i) The correct statement regarding Freundlich isotherm regarding physical adsorption is $\left(\frac{x}{m}\right) = kP^{\circ}$ **at high pressure.**

This is because at high pressure, the entire area of the adsorbate is completely covered by adsorbent.

(ii) $\log\left(\frac{x}{m}\right) = \log k + \frac{1}{n}\log p$

Substitution gives $\log \frac{16.5}{3} = \log \mathrm{k} + \frac{1}{\mathrm{n}} \log 180$

$\log \frac{38.1}{3} = \log \mathrm{k} + \frac{1}{\mathrm{n}} \log 540$

On subtraction, $\log \frac{38.1}{16.5} = \frac{1}{\mathrm{n}} \log \frac{540}{580}$

$\log 2.3091 = \frac{1}{\mathrm{n}} \log 3$ **or** $\frac{0.4771}{0.3631} = 1.31$

Q.17 Answer:

$0.1044 \mathrm{ccg}^{-1}$

Solution:

(iii) $\left(\frac{x}{m}\right) = kp^{1/n}; \log \frac{x}{m} = \log k + \frac{1}{n}\log p$

$\log \frac{16.5}{3} = \log \mathrm{k} + \frac{1}{1.31} \log 180; \log 5.5 = \log \mathrm{k} + \frac{2.2553}{1.31}$

$\log \mathrm{k} = \log 5.5 - \frac{2.2553}{1.31} = 0.7404 - 1.7216 = -0.9812 = \overline{1}.0198$

$\mathrm{k} = 0.1044 \mathrm{~ccg}^{-1}$

Q.18 Answer:

$(\mathrm{A}) \rightarrow \mathrm{p}, (\mathrm{B}) \rightarrow \mathrm{q}, \mathrm{r}, (\mathrm{C}) \rightarrow \mathrm{q}, \mathrm{s}, (\mathrm{D}) \rightarrow \mathrm{r}$

Solution:

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$(A) - p, (B) - qr, (C) - qs, (D) - r$

(A) Solution-homogeneous

(B) Colloid - Heterogenous shows Tyndall effect

(C) Suspension - Heterogenous affected by gravitational force

(D) Optical property - Tyndall effect

Q.19 Answer:

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Solution:

Mass of acetic acid adsorbed by 2 **g charcoal** $= 100 \times 10^{-3} \times (0.6 - 0.5) \times 60 = 0.6$

$\Rightarrow \frac{x}{m} = \frac{0.6}{2} = 0.3$

Q.20 Answer:

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Solution:

Adsorbed moles of $H_2 = \frac{0.03 \times 2.46 \times 10^{-3}}{0.0821 \times 300} = 3 \times 10^{-6}$

$\therefore$ **No. of absorbed molecules of** $H_2 = 3 \times 10^{-6} \times 6 \times 10^{23} = 18 \times 10^{17}$

Total no. of surface sites available $= 6 \times 10^5 \times 1000 = 6 \times 10^{18}\ cm^2$

No. of surface sites that is occupied by adsorption of

$H_2 = \frac{10}{100} \times 6 \times 10^{18} = 6 \times 10^{17}$

No. of surface sites occupied by one molecule

$H_2 = \frac{18 \times 10^{17}}{6 \times 10^{17}} = 3$

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